

1.Import libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
```

2.Load/Create dataset

```
data = {
    'hours_studied': [1, 2, 3, 4, 5, 6, 7, 8],
    'marks': [35, 40, 50, 55, 65, 70, 80, 90]
}

df = pd.DataFrame(data)
print(df)
```

	hours_studied	marks
0	1	35
1	2	40
2	3	50
3	4	55
4	5	65
5	6	70
6	7	80
7	8	90

3.Define features (X) and target (y)

```
X = df[['hours_studied']] # Input (Independent variable)
y = df['marks']           # Output (Dependent variable)
```

4.Split data into training and testing sets

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

5.Create Linear Regression model , Train (fit) the model

```
model = LinearRegression()
model.fit(X_train, y_train)
```

▼ LinearRegression ⓘ ?

```
LinearRegression()
```

6. Make predictions

```
y_pred = model.predict(X_test)

print("Predictions:", y_pred)
print("Actual:", y_test.values)
```

```
Predictions: [41.7 72.9]
Actual: [40 70]
```

7. Make predictions, Evaluate the model

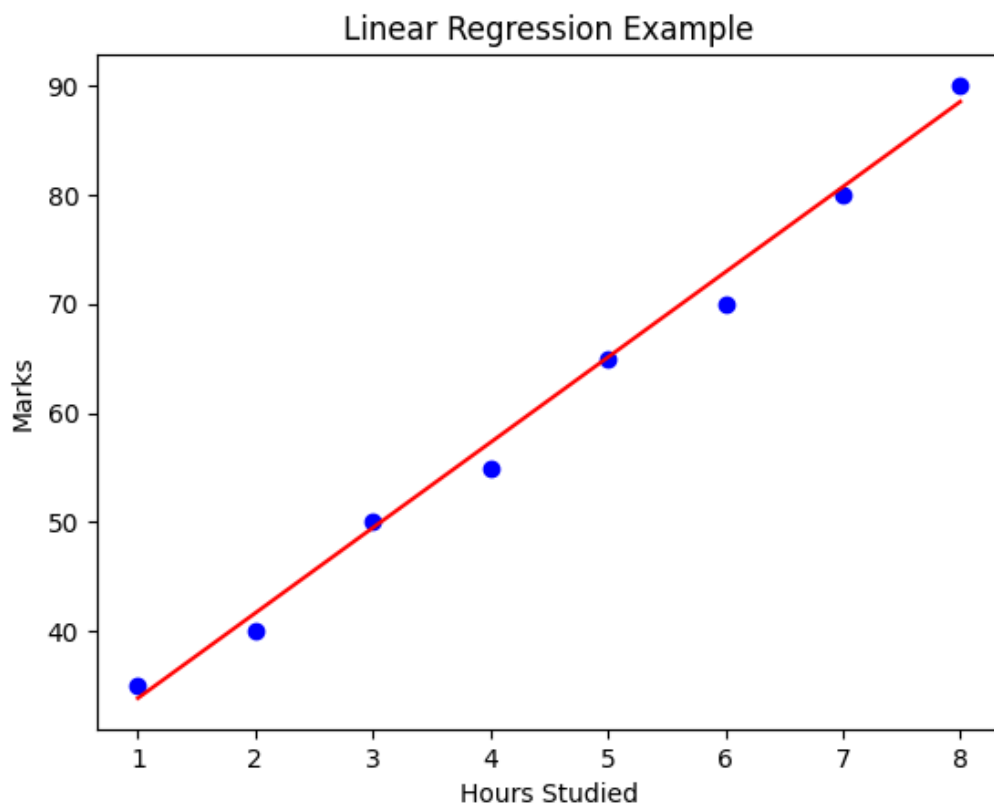
```
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Absolute Error:", mae)
print("R2 Score:", r2)
```

```
Mean Absolute Error: 2.3000000000000007
R2 Score: 0.9748888888888888
```

Visualize results (optional) Predict new values

```
plt.scatter(X, y, color='blue')          # actual data
plt.plot(X, model.predict(X), color='red') # regression line
plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.title("Linear Regression Example")
plt.show()
```



```
new_hours = np.array([[6]])  
predicted_marks = model.predict(new_hours)  
  
print("Predicted Marks for 9 hours:", predicted_marks[0])
```

```
Predicted Marks for 9 hours: 72.9  
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning:  
  warnings.warn(
```